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PAPER

Surface pattern formation in a swelling gel

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We study the time evolution of surface pattern formation in a swelling spherical gel using proton nuclear magnetic resonance microimaging of water molecules migrating into the gel. Relaxation-weighted images permit mapping of the local evolution of the swelling ratio which, within the framework of a nonlinear poroelastic theory, can be identified with the determinant of the deformation gradient tensor. The wavelength of the surface pattern is examined and some analogies with the extensively studied problem of the wavelength at threshold for swelling rings are pointed out. By controlling the degree of cross-linking within a surface layer of the gel it is possible to monitor the transition from an undulating folding pattern to the formation of singular creases.

Introduction

The problem of heterogeneous growth in layered structures has attracted considerable interest after it was realized that it plays an important role in the morphogenesis of many biological structures such as: fingerprints,¹ brain convolutions,² and the shape of some fruits.³ Moreover, the shape of some tumors, such as melanoma, was also found to be associated with heterogeneous growth.⁴ Recently, a model consisting of a swelling ring of gel, with its inner circle constrained to the perimeter of a non-swelling gel disk, was proposed as an artificial tumor.⁵ Most importantly the predictions of a nonlinear poroelastic theory¹¹ concerning surface pattern formation in this system were confirmed.

In their pioneering work Tanaka *et al.*^{6,7} modeled the time evolution to equilibrium of a swelling gel using a kinetic equation involving a friction coefficient between polymer and solvent and Hookean elasticity theory. This approach was useful to describe the relatively small amount of swelling of neutral polyacrylamide spheres in water. However for ionic polyacrylamide copolymers, where the substantial osmotic pressure exerted by the counterions produces a large degree of swelling, nonlinear effects become important and a linear theory based upon Hooke's law breaks down.^{8,9} Furthermore, the behavior of ionic polyacrylamide gels in water, where the degree of swelling is large, was found to exhibit a remarkable property.¹⁰ As the swelling progresses a regular pattern develops on the surface of the gel which breaks the original symmetry. The characteristic size of the pattern units grows until it achieves an equilibrium size determined by external mechanical constraints. In the case of a sphere, without any constraints, the size of the surface pattern grows until it becomes

large compared to the surface of the sphere and the initial spherical symmetry is restored. The origin of these patterns was attributed to an inhomogeneous swelling process where the surface is fully swollen but the innermost layers are not. Moreover, the average wavelength of the surface pattern was found to be proportional, as predicted by a stability analysis,¹⁰ to the total thickness of the layered structure.

Another aspect of the formation of patterns on the surface of swelling gels has gained much attention recently.¹¹ The surface pattern units are known to often be separated by highly singular creases^{5,12,13} or sulci which are expected to form on soft matter as a consequence of a compression-induced instability.¹³ In the case of swelling gel rings the ratio of Young's modulus of the external ring to that of the non-swelling internal gel disks was found to play an important role in determining the onset of either sulcification or folding undulations in the surface pattern.⁵

In NMR, spin-lattice and spin-spin relaxation rates are strongly dependent on the tumbling motion of water molecules and can be conveniently incorporated into the images to provide contrast. Thus image intensity becomes sensitive not only to the local water content but also to the mobility of the water molecules.¹⁴ From such images it is possible to infer, with acceptable time resolution, the kinetics of the local degree of swelling and its relationship to the dynamics of surface pattern formation. Furthermore, using small amounts of Cu²⁺ ions as a contrast agent it is possible to monitor, with good spatial resolution, the formation of sulci and undulations.

In this paper we demonstrate that high magnetic field nuclear magnetic resonance microimaging can be a powerful tool to study the kinetics of swelling and surface pattern formation in gels.

Materials and methods

Commercial ionic polyacrylamide copolymer beads of different diameters and stiffness were employed. Equilibrium swelling

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ratios in the range $1 < J_0 < 225$ could be achieved under various swelling conditions, where $J_0 = 1/\phi_0$ and ϕ_0 denotes the equilibrium volume fraction of polymer. Elastic moduli values were estimated by measuring the deformation of swollen spheres subjected to a uniaxial compression by a contact load against a non-deformable flat.¹⁵ This yields Hertz's elastic modulus E^* which, for a non-deformable flat, is related to Young's modulus E and Poisson's ratio ν of the gel by $1/E^* = (1 - \nu^2)/E$. Moreover, the shear modulus is given by $G = E/2(1 + \nu)$.

To characterize the swelling pressure and compare with other ionic polyacrylamide gels^{16,17} the variation of the equilibrium swelling ratio J_0 of the beads with the ionic strength of the solutions was measured for a monovalent cation (Na^+) and for bivalent cations (Cu^{2+} and Ca^{2+}).

Nuclear magnetic resonance (NMR) measurements were performed in a Varian MR System with VnmrJ 2.3A software. The system operates at a proton frequency of 400 MHz corresponding to a magnetic field of 9.39 T and the probe has a useful diameter of 20 mm. Spin-echo multiple slice sequences¹⁴ were employed for all images with a slice thickness of 0.5 mm and image matrix size of 512×512 and an approximate voxel size of $0.05 \times 0.05 \times 0.5 \text{ mm}^3$.

Background and analysis

Although a full mathematical description of pattern formation in a swelling sphere is beyond the scope of this paper it is useful to introduce some basic concepts to facilitate the discussion of our results. Hong *et al.*⁸ have recently developed a nonlinear poroelastic kinetic theory. It describes coupled diffusion and large deformation in polymeric gels. The theory has two material specific assumptions: the free-energy function, for which the Flory–Huggins¹⁸ form is adopted, and the kinetic law which assumes that the small solvent molecules diffuse inside the gel.

The dry polymer network under no mechanical load is taken as the reference state and X_α ($\alpha = 1, 2, 3$) denotes the coordinate of a marker. In the current state at time t the marker has moved to a point with a coordinate $x_\beta(\vec{X}, t)$. The deformation gradient tensor $F_{\beta\alpha}$ is defined by:

$$F_{\beta\alpha} = \frac{\partial x_\beta(\vec{X}, t)}{\partial X_\alpha} \quad (1)$$

and the determinant of $\vec{F}$ plays a special role. Since the long polymer molecules as well as the small solvent molecules are not expected to change volume and no voids are expected to be present, a condition of molecular incompressibility can be invoked. This can be expressed by:⁸

$$1 + \nu C = \det(\vec{F}), \quad (2)$$

where ν is the molecular volume of a solvent molecule and νC denotes the volume element occupied by the solvent molecules in the current state divided by the volume of dry polymer. Hence the degree of swelling, or dilatation, is given by $\det(\vec{F}) = J(\vec{X}, t) = 1/\phi(\vec{X}, t)$, where $\phi(\vec{X}, t)$ denotes the volume fraction of polymer in the current state.

Defining the symmetrical tensor $(\vec{F})^T \vec{F}$ and denoting its eigenvalues as λ_1^2 , λ_2^2 , and λ_3^2 the determinant of the deformation gradient tensor becomes $\det(\vec{F}) = J = \lambda_1 \lambda_2 \lambda_3$ where λ_α ($\alpha = 1,$

2, 3) are the three positive principal stretches. From the Flory–Huggins¹⁸ free-energy function $W(C, \vec{F})$, together with the microscopic incompressibility condition of eqn (2), and a kinetic equation relating the flux of solvent particles to the gradient of the chemical potential it is possible to derive a set of nonlinear differential equations⁸ for the principal stretches $\lambda_\alpha(\vec{X}, t)$. However, this approach alone does not address the problem of pattern formation in a swelling gel since such patterns arise from the growth of unstable modes as a response to a small periodic perturbation. This aspect of the problem requires a stability analysis of the equations.

For the case of a ring of swelling gel with its inner circumference constrained to the perimeter of a non-swelling disk Dervaux *et al.*^{19,5} developed a linear stability analysis. Denoting by A the radius of the non-swelling disk, H the annular width of the ring in the reference state and h the annular width in the current state, a threshold value of h/H for the onset of the instability was predicted.¹⁹ The measured wavelength ratio λ/H at threshold was found⁵ to be approximately constant for $H/A < 0.4$. Furthermore as long as $E_d/E_r > 1$, where E_d and E_r are Young's moduli of disk and ring, λ/H was also found to be weakly dependent upon E_d/E_r , as predicted by the linear stability analysis.⁵

NMR microimaging

The NMR signal is generated by the Larmor precession of the magnetic moments of water protons in a magnetic field. After the encoding of spatial information through the application of magnetic field gradients, an image can be reconstructed representing a proton density map weighted by the relaxation times.¹⁴ In a conventional saturation recovery spin-echo image²⁰ the NMR signal amplitude S from a voxel containing water and polymer is given by:²⁰

$$S(\phi) = S_0(1 - \phi)\exp(-T_E/T_2)(1 - \exp(-T_R/T_1)) \quad (3)$$

where ϕ denotes the volume fraction of polymer in the voxel and T_1 and T_2 are, respectively, spin–lattice and spin–spin relaxation times. T_E and T_R are operationally adjustable and represent, respectively, the echo time and the repetition time of the sequence. Furthermore, provided the conditions $T_E \ll T_2$ and $T_R \gg T_1$ are satisfied, S_0 represents the expected signal from pure water ($\phi = 0$), since no signal from protons in the polymer can be detected under our experimental conditions.

Although eqn (3) correctly describes the signal amplitude for $T_R > T_2$, when the repetition time becomes comparable or smaller than T_2 , the factor $(1 - \exp(-T_R/T_1))$ in eqn (3) must be modified. In this situation the condition leading to a steady state free precession regime^{21,22} is approached. Hence, for $\pi/2$ pulses and zero offset between the excitation frequency and the Larmor frequency, an extra multiplicative correction factor $1/(1 + \exp(-T_R/T_2)\exp(-T_R/T_1))$ must be included in eqn (3).

In principle, by making $T_E \ll T_2$ and $T_R \gg T_1$ one could obtain the volume fraction of polymer ϕ from eqn (3) by measuring $S(\phi)/S_0$.

In practice, however, there are several limitations. If one attempts to make $T_R \gg T_1$ the imaging time becomes prohibitively long on the time scale of the swelling process. Furthermore,

there is a limit on how short T_E can be made for a given field of view and slice thickness.

There is another reason, besides the time scale of the process, to make T_R shorter than T_1 in eqn (3). The factor $(1 - \phi)$ alone is not able to provide sufficient contrast in regions where the dilatation $1/\phi$ is larger than typically 5. The choice of T_R therefore becomes a compromise between better contrast and satisfactory signal to noise ratio. To obtain the local dilatation $J = 1/\phi$ in a swelling gel with acceptable time resolution and image contrast from measurements of the image amplitude S/S_0 it is therefore necessary to determine the ϕ dependence of the relaxation times and find the solutions of eqn (3) for a range of values of S/S_0 .

From a microscopic point of view a proton in a free water molecule can hydrogen-bond to a polymer chain for a brief period of time and later become free again. Since this exchange process takes place in a time scale much shorter than typical NMR relaxation times the measured relaxation rates $R_{1,2}$ can be expressed simply as weighted averages²³ over the two states:

$$\frac{1}{T_{1,2}} = R_{1,2} = \left(\frac{1}{T_{1,2}}\right)_b p_b + \left(\frac{1}{T_{1,2}}\right)_f (1 - p_b), \quad (4)$$

where $(1/T_{1,2})_b$ and $(1/T_{1,2})_f$ respectively denote proton relaxation rates for bound and free water molecules and p_b is the probability to find a bound water molecule. If one assumes²³ that the probability p_b is simply given by the volume fraction of polymer, and $p_b = \phi$ is inserted in eqn (4), the relaxation rates become linear functions of ϕ .

Eqn (4) was tested experimentally in the following manner. Several polymer beads were permitted to swell in slightly doped water solution for periods of time varying between 1 min and 60 min. Following the swelling each bead was removed from the bath and allowed to equilibrate in air for approximately two hours. After this waiting time, all evidence of surface patterns had disappeared and the spherical shapes were recovered with larger diameters corresponding to the longer swelling times. NMR images of these equilibrated spheres revealed that the dilatation was spatially uniform. Furthermore, since no observable water signal was detected in the dry polymer beads the volume fraction of polymer in each bead can be assumed to be given by $\phi = (A/b)^3$, where b is the radius of the swollen bead and A denotes the radius of the dry polymer bead. Finally, using conventional NMR methods, T_1 and T_2 were measured on all the spheres as a function of ϕ .

The validity of the assumptions leading to eqn (4) could be questionable if, in addition to water protons and slowly moving polymer chains, a sizable amount of mobile monomers were present. No appreciable signal from protons in mobile monomers was observed in the samples employed here. However, in some other types of high-swelling gels, a proton signal from mobile monomers was detectable in the dry polymer using $T_E \sim 20$ ms. In principle, the method described here could still be employed even in these cases, albeit with a possibly more complex relationship than that of eqn (4), which could be determined experimentally.

Experimental results

All measurements were performed on samples of ionic polyacrylamide copolymer beads. Samples 1 and 2 differed only in

their initial dry diameters. Sample 1 had a diameter $2A = 3.15$ mm in the reference state and sample 2 had a diameter $2A = 2.5$ mm. The measured Hertz elastic modulus at $J = 1/\phi = 20$ was $E^* = 84$ kPa, this corresponds, assuming a typical value of $\nu = 0.35$ for Poisson's ratio,²⁴ to a Young's modulus $E = 74$ kPa and a shear modulus $G = 27$ kPa.

The equilibrium dilatation of samples 1 and 2 in Milli-Q water was $J_0 = 225$ and dropped to $J_0 = 68$ at a NaCl concentration of 80 mM. Moreover the effect of divalent cations was quite dramatic, as observed earlier in Na-polyacrylate.^{16,17} For Ca^{2+} and Cu^{2+} a concentration of only 4 mM was sufficient to lower the equilibrium dilatation to $J_0 = 9$.

Fig. 1 shows measurements of spin-lattice and spin-spin relaxation rates in eight different spheres of sample 1 with various degrees of swelling.

From a linear fit to the data of Fig. 1, one obtains the following values for the parameters of eqn (4): $(1/T_1)_f = 0.36 \text{ s}^{-1}$, $(1/T_1)_b = 4.46 \text{ s}^{-1}$ and $(1/T_2)_f = 0.70 \text{ s}^{-1}$, $(1/T_2)_b = 32.2 \text{ s}^{-1}$.

The images in Fig. 2 illustrate with a few examples the evolution of the swelling process for ionic polyacrylamide beads (sample 1) at times $t = 5, 10, 18$, and 26 min after the initiation of swelling. The sagittal images shown in Fig. 2 were obtained using a spin-echo multiple slice sequence with $T_E = 22$ ms and $T_R = 500$ ms. Five slices, each with a 0.5 mm width, were acquired but only those which most closely coincide with a central plane are displayed. The size of the data matrix was 512×512 with a voxel size $0.05 \times 0.05 \times 0.5 \text{ mm}^3$ and the imaging time was approximately 4 min.

A spherical bead with an initial diameter $2A = 3.15$ mm was introduced at $t = 0$ in a 20 mm diameter cylindrical tube containing a 0.1 mM solution of CuSO_4 . The dilute solution of paramagnetic Cu^{2+} ions shortens T_1 permitting a good contrast between the surface pattern and the surrounding fluid. By comparing these images with the images obtained in pure water we conclude that T_1 is only affected in a thin layer at the interface. Furthermore, the equilibrium dilatation J_0 is weakly affected by the presence of the Cu^{2+} ions at this low concentration.

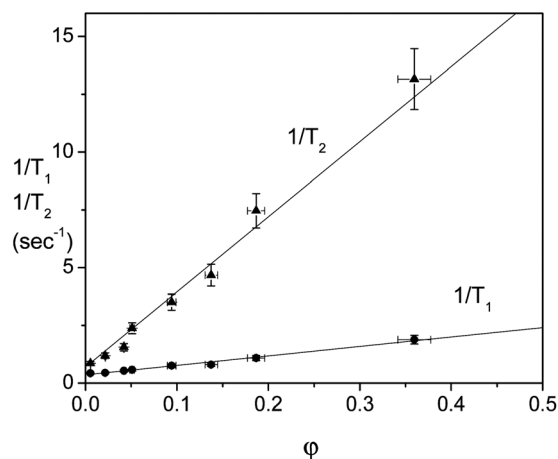


Fig. 1 (●) Spin-lattice and (▲) spin-spin ^1H relaxation rates as a function of volume fraction of polymer ϕ for eight spheres of sample 1 with different degrees of swelling.

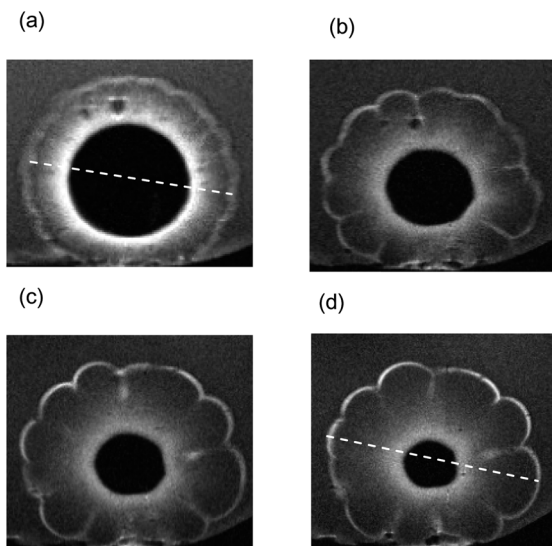


Fig. 2 Sagittal 0.5 mm slice of an initially spherical ionic polyacrylamide sample (sample 1) immersed in a 0.1 mM CuSO_4 solution. Dashed segments indicate maximum diameters of lengths $2b$. (a) $2b = 6$ mm at time $t = 5$ min. (b) $2b = 7$ mm at $t = 10$ min. (c) $2b = 7.5$ mm at $t = 18$ min, and (d) $2b = 8.2$ mm at $t = 26$ min after immersion.

Fig. 3 shows similar images to those in Fig. 2 for longer times: $t = 33, 57, 65$ and 81 min after immersion. The surface patterns in Fig. 2 and 3 clearly exhibit the characteristic petals separated by singular creases.

From the images of Fig. 2 and 3 normalized signal amplitude profiles S/S_0 , where S_0 denotes the water signal amplitude in the surrounding bath measured under the conditions $T_R \gg T_1$ and $T_E \ll T_2$, can be obtained. In Fig. 4 such profiles were obtained for the representative images of Fig. 2a and d and 3b and c along ranges, such as those shown by the dashed lines of Fig. 2 and 3, which do not include any creases.

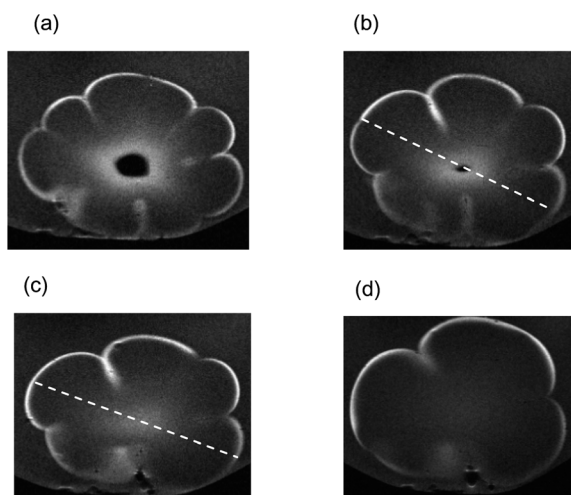


Fig. 3 Sagittal 0.5 mm slice of an initially spherical ionic polyacrylamide sample (sample 1) immersed in a 0.1 mM CuSO_4 solution. Dashed segments indicate maximum diameters of lengths $2b$. (a) $2b = 9.5$ mm at time $t = 33$ min. (b) $2b = 10.5$ mm at $t = 57$ min. (c) $2b = 11.5$ mm at $t = 65$ min and (d) $2b = 12$ mm at $t = 81$ min after immersion.

Fig. 4 also shows, for each measured profile, calculated values of the dilatation $J = 1/\phi$ obtained from eqn (4) and eqn (3) with its small correction factor. Also shown are the data of Fig. 1.

Discussion

Although NMR imaging has been successfully employed in the study of transport properties in hydrogels,^{25–28} the relationship between the kinetics of surface pattern formation and the spatial and temporal dependence of the dilatation has not been previously addressed using this technique. The dynamics of swelling in thin gel layers, however, has been recently studied, using

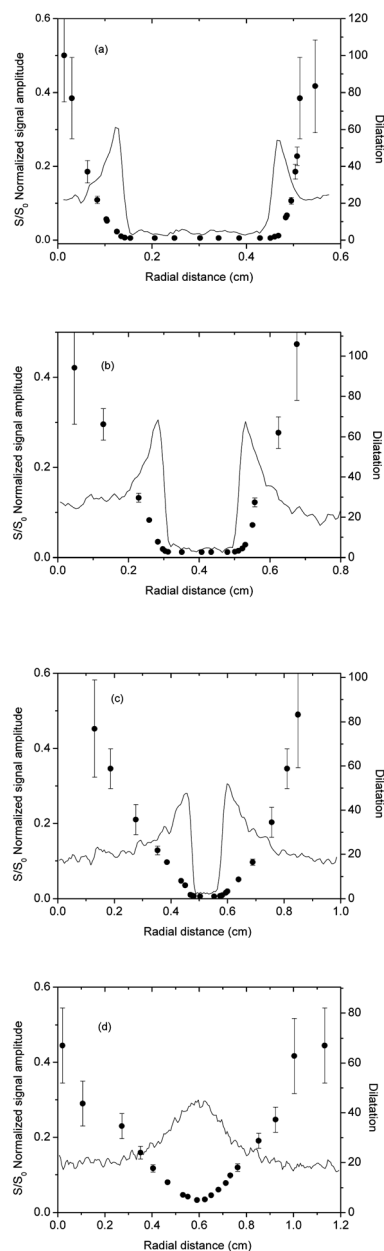


Fig. 4 (—) Normalized signal amplitude (left vertical scale) and (●) calculated values of dilatation J (right vertical scale) along a radial range such as shown by the dashed lines in Fig. 2 and 3. (a) For image 2a. (b) For image 2d. (c) For image 3b. (d) For image 3c.

fluorescence spectroscopy, by monitoring the time evolution of the positions of micron-size fluorescent polystyrene beads embedded in the gel.²⁹ The method presented here has the advantage of not requiring the presence of an external probe and of yielding directly the dilatation J .

The inspection of Fig. 2–4 immediately leads to quantitative conclusions. During most of the process there exists a core of diminishing diameter, which does not swell at all, where the dilatation has a value $J = 1$. Outside this core, shown as black circles with $S/S_0 = 0$ in Fig. 2 and 3, one observes a bright ring where the signal has the highest value, as shown in the profiles of Fig. 4a–c. This signal intensity is not caused by a maximum in water content but rather by the product of the term $1 - \phi$ in eqn (3), which decreases with increasing ϕ , and the relaxation term $1 - \exp(-T_R/T_1)$ which increases with ϕ . From the profiles of Fig. 4 the values of the dilatation at the signal maximum are found to be $J \approx 6$ and they rapidly increase to values close to $J \approx 100$ as the boundary of the petal is approached.

Unfortunately, given the available contrast, the signal amplitude in this highly swollen ($\phi \sim 0.01$) region, as given by eqn (3), becomes rather insensitive to ϕ and the error in the measured values of J becomes large.

It is worth noticing, as suggested by Fig. 2 and 3, that the broken spherical symmetry caused by the instability of some modes of perturbation persists even when only relatively minor dilatation gradients are present. In Fig. 3d, for example, although a variation in signal amplitude is barely detectable, a surface pattern consisting of three petals is still observed.

The existence of a regime with a non-swollen core of diminishing current radius a , where the condition $J = 1$ prevails, permits a comparison with the problem of a swelling annular ring attached to a non-swelling disk where the wavelength of the surface pattern at threshold has been studied extensively.^{5,19} In the present case one could define the value $h = b - a$ as the current width of the annular ring obtained from the imaging slice. Here b denotes the current radius measured from the center of the bead to the edge of the petal, as defined in Fig. 2 and 3, and a is the current radius of the $J = 1$ region where $S/S_0 = 0$. Since h increases from the value $h = 0$ in the reference state, the wavelength is expected to grow from zero reaching successive thresholds¹⁹ for modes with increasingly larger wavelengths. Although more than a single mode could be present at a given surface pattern one could still try to determine the dependence of the average wavelength λ_{av} with h . We consequently define¹⁹ $\lambda_{av} = 2\pi A/m$, where m denotes the number of petals in an image slice such as shown in Fig. 2 and 3.

Fig. 5 shows a plot of $\lambda_{av}/A = 2\pi/m$ as a function of $h/A = (b - a)/A$ for a variety of samples with different diameters in the reference state. In addition to samples 1 and 2, sample 3 has a dry diameter $2A = 1.5$ mm and sample 4 has $2A = 3.5$ mm. These two samples have different stiffness from samples 1 and 2.

Fig. 5 indicates that for $(b - a)/A < 3.5$ an approximately linear relationship exists between $\lambda_{av}/A = 2\pi/m$ and $(b - a)/A$ with a slope $\lambda_{av}/(b - a) = 0.33 \pm 0.03$ which is weakly dependent upon the stiffness of the dry polymer. This parallels the behavior^{5,19} of thick rings where a slope 0.4 has been predicted for $(B - A)/A = 1$. It should be pointed out however that in the case of thick rings B and A refer to radii in the reference state whereas b and a in Fig. 5 are continuously varying radii in the current state.

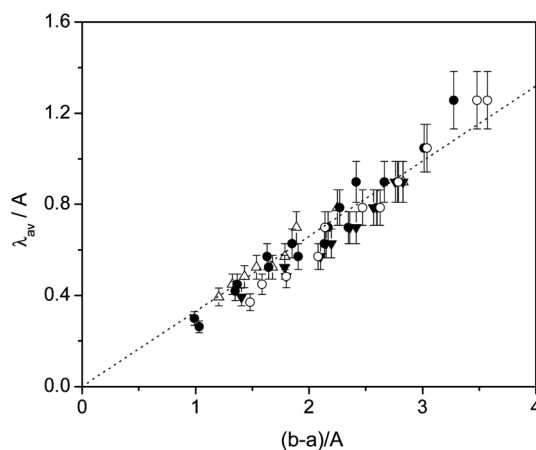


Fig. 5 Average wavelength λ_{av} as a function of current thickness $(b - a)$ for four different gel samples: (●) sample 1; (▼) sample 2; (○) sample 3; and (Δ) sample 4.

Another important aspect of the surface patterns is the highly singular creases separating the petals that can be observed in Fig. 2 and 3. In the case of swelling rings it has been shown⁵ that if the stiffness of the annular ring becomes larger than the stiffness of the inner disk a transition from creases to undulations takes place. In the present problem of surface patterns on a swelling sphere we have a different situation; we can control the transition from undulations to singular creases by reducing the stiffness of a thin surface layer¹³ via cross-linking Cu^{2+} ions.

Measurements of the shear modulus showed^{16,17} that the dramatic drop in the equilibrium dilatation caused by divalent cations such as Ca^{2+} on sodium polyacrylate gels could not be attributed to an increase in cross-linking. This is consistent with our observation that Ca^{2+} ions are not able to cause an evolution from creases to undulations. To study the effect of Ca^{2+} ions we imaged the surface patterns of spheres swelling in a 5 mM solution of CaCl_2 . Although the equilibrium dilatation was reduced drastically the singular creases between petals were still present and were not affected by the presence of Ca^{2+} .

A completely different behavior was found using Cu^{2+} ions which appear to become localized on a thin surface layer and visibly increase the stiffness. The ability of Cu^{2+} ions to form complexes with ligands attached to the network of acrylic acid–polyacrylamide copolymers has been reported earlier.³⁰ When the stability of the complex is high, the complexation can be regarded as a formation of new cross-linkages which lead to an increase in stiffness. The theory of rubber elasticity¹⁸ predicts a value $G \sim kT\nu\phi^{1/3}$ for the shear modulus. Here ν represents the number of segments along a primary molecule extending from one cross-linkage to the next per unit volume of dry polymer, k denotes Boltzmann's constant, T absolute temperature, and ϕ is the volume fraction of polymer. Each new cross-linkage increases the number of segments by one unit and the shear modulus increases by approximately $kT\phi^{1/3}/V$, where V is the volume of dry polymer in the gel. Moreover, in the case of Ca^{2+} , aggregates also appear to form but, since the elastic properties are not affected, they appear to be rather loose.¹⁶

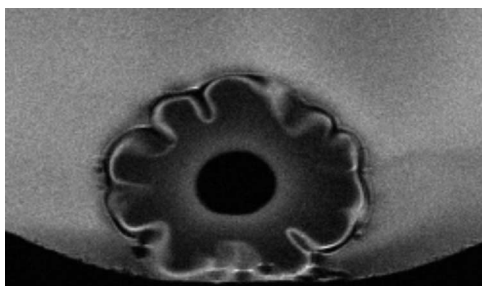


Fig. 6 Surface folding undulations in a sagittal 0.5 mm slice of a sample 1 gel after 30 minutes immersion in a 5 mM CuSO_4 solution. The diameter of the dry polymer region is $2a = 1.96$ mm.

Fig. 6 shows a sagittal image obtained under the same conditions as Fig. 2 and 3 for a sphere of sample 1 immersed in a 20 mm diameter cylindrical tube containing a 5 mM CuSO_4 solution. The surface pattern exhibits folding undulations but no longer displays the singular creases apparent in Fig. 2 and 3. The transition from creasing to folding was found to take place in the range of concentration 1.5–2 mM.

Conclusions

Proton NMR microimages of water migrating into a spherical gel bead permit determination of the spatial and temporal dependence of the dilatation J during the evolution of the mechanical instability which leads to surface patterns. This quantity, which within the framework of a non-linear poroelastic theory is given by the determinant of the gradient deformation tensor, is expected to be of central importance for a comparison between theory and experiment. The images reveal that, before gradients of dilatation have vanished and spherical symmetry is restored, there exists a central core of diminishing diameter where the condition $J = 1$ holds for a considerable time interval.

This establishes a surprising parallel between the behavior of a slice of swelling spherical gel and a swelling annular ring attached to a non-swelling inner disk. For four different samples with different diameters and stiffness a value $\lambda_{av}/(b - a) = 0.33 \pm 0.03$ was found for $(b - a)/A < 3.5$. The quantities b and a respectively denote the current largest radius of the swelling bead and the current radius of the $J = 1$ region, whereas A denotes the radius of the bead in the reference state. The average wavelength is given by $\lambda_{av} = 2\pi A/m$ where m denotes the number of petals.

Finally, in our experiments, the transition from singular creases in the surface pattern to folding undulations can be controlled by creating a thin layer of stiffer gel via the cross-linking effect of Cu^{2+} ions.

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